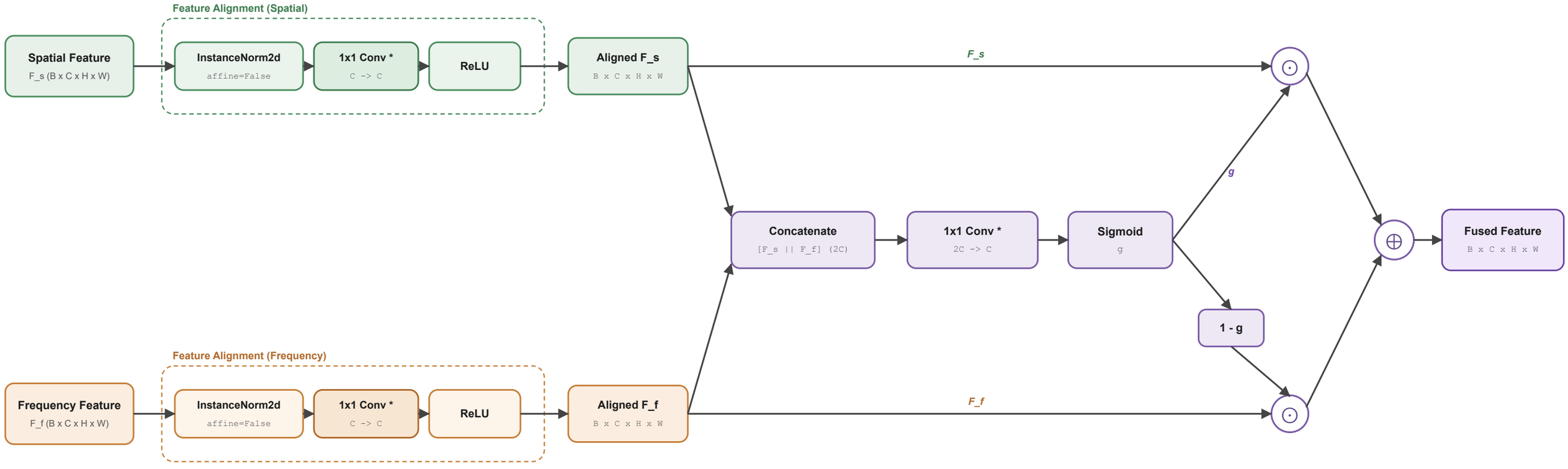
Feature Alignment and Gated Fusion of the Proposed MAFF-Net



Gate: $g = \sigma(\text{1x1 Conv}([F_s \,||\, F_f]))$ Fusion: $F_{fused} = g \odot F_s + (1 - g) \odot F_f$

*** learnable (trainable parameters). InstanceNorm2d uses affine=False (parameter-free); ReLU, Sigmoid, concatenation, ⊙ (element-wise multiply) and ⊕ (element-wise add) are parameter-free.**

Spatial path Frequency path Gated Fusion ⊙ element-wise multiply ⊕ element-wise add [· || ·] channel concatenation